Uncovering the origin of malaria in the Americas using Deep Learning approaches on genomic data

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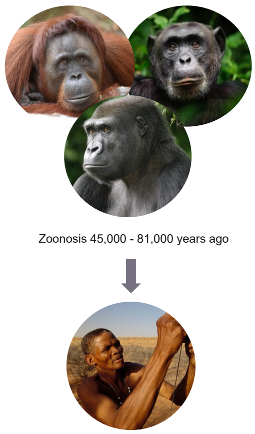
1. INTRODUCTION

Malaria, a parasitic disease endemic to most tropical and subtropical regions, remains one of the most severe life-threatening **infectious diseases** worldwide. Beyond its contemporary health burden, malaria has also been a major driver of human evolution, representing one of the strongest selective pressures identified in the **human genome**. Despite its importance, the demographic history of *Plasmodium vivax*—one of the major human malaria parasites—in the Americas remains poorly understood (**Figure 1**). Here, we evaluate whether alternative demographic scenarios for the introduction of P. vivax into the Americas can be distinguished using a multilayer perceptron (MLP) machine-learning classifier.

A)



B)

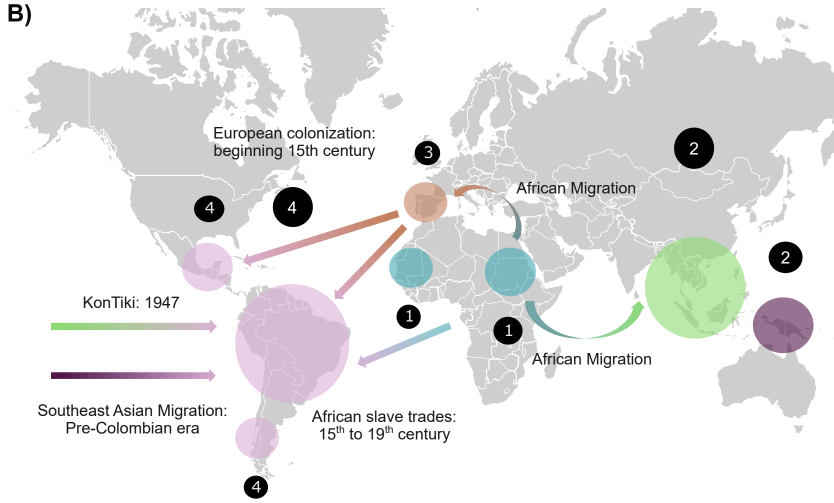


Figure 1. A) Possible zoonotic origin events. B) Hypotheses for the introduction of *Plasmodium vivax* into the Americas. Map generated with information from Rodrigues et al. 2018

2. METHODS

1. Building demographic models (**Figure 2**) in **msprime**

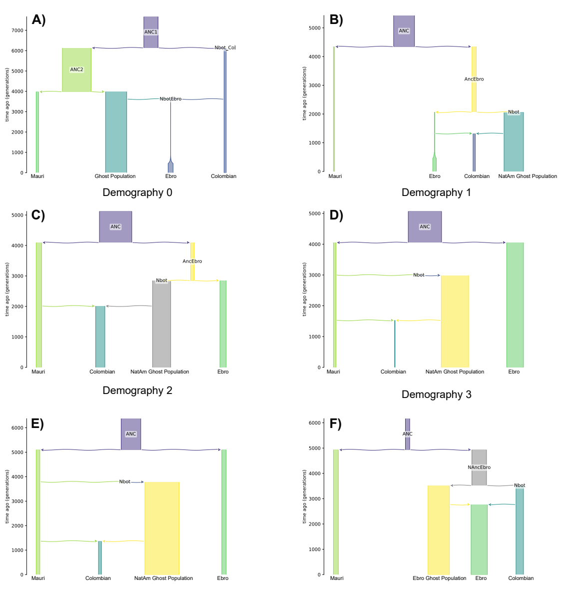


Figure 2. Multiple hypotheses for *P.vivax* introduction in the Americas. Models modified from Lefebvre et al. 2025

2. Simulate genomic data for *P. vivax* and calculate **summary statistics (SS)**

3. Train a MLP machine learning classifier using SS

3. RESULTS

A)

B)

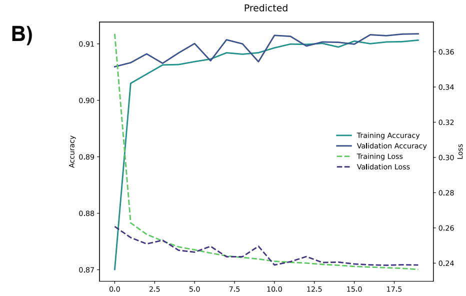


Figure 3. MLP Classifier results. A) Confusion Matrix. B) Loss and accuracy history

4. CONCLUSION

Machine learning algorithms can accurately (Accuracy: 0.90) classify and distinguish different demographic scenarios for *P. vivax* arrival in the Americas.

References: Rodrigues PT, Valdivia HO, de Oliveira TC, Alves JMP, Duarte AMRC, Cerutti-Junior C, et al. Human migration and the spread of malaria parasites to the New World. Sci Rep. 2018 Jan 31;8(1):1993. González-Sanz M, Berzosa P, Norman FF. Updates on Malaria Epidemiology and Prevention Strategies. Curr Infect Dis Rep. 2023 Jul 1;25(7):131–9. Lefebvre MJM, Degrugillier F, Arnathau C, Fontecha GA, Noya O, Houzé S, et al. Genomic exploration of the journey of Plasmodium vivax in Latin America. PLOS Pathog. 2025 ene;21(1):e1012811.